

YOUR RESPONSIBILITIES vs. MY CODE

October 2, 2026 → End of February 2027

Dissertation: MSc Class Imbalance Handling in Financial Fraud Detection

Submission Deadline: End of February 2027

Total Timeline: 23 weeks (Oct 2 - Feb 28)

Your Total Time Investment: 70.5 hours

WEEK 10-11 (Oct 2-16) — LATENCY PROFILING & RESULTS ANALYSIS

What I Provide:

`src/inference_profiler.py` (150+ lines) - Load all 12 trained models from MLflow - Measure inference latency per technique (mean, std, 95th percentile) - Profile memory usage, CPU utilization - Test across 100, 1,000, 10,000 transaction batches - Export timing data to CSV

`notebooks/03_results_analysis.ipynb` - PR curves for top 5 techniques (plotly interactive) - Confusion matrices (heatmaps) - Feature importance plots (SHAP values on final model) - Error analysis: false positives vs false negatives by threshold - **Accuracy vs Latency scatter plot** (your novel contribution visualization) - Operating point recommendations at different thresholds - Friedman-Nemenyi rankings table - Statistical significance annotations

What You Do:

Task	Time	Output
Review <code>inference_profiler.py</code> (skim code, understand flow)	15 min	Notes on any clarifications needed
Run <code>inference_profiler.py</code> on your hardware	45 min	<code>latency_results.csv</code> with all timing data
Execute <code>notebooks/03_results_analysis.ipynb</code> cell-by-cell	1 hour	All plots generated, rankings confirmed
Write Chapter 5 §5.4 (Latency Analysis)	2 hours	800-1000 words + Figure 8 (latency plot)
Write Chapter 5 §5.5 (Error Analysis)	2 hours	800-1000 words + confusion matrices + threshold discussion
Week 10-11 Total	5.75 hours	180 models analyzed, latency-accuracy trade-off documented

WEEK 12-13 (Oct 16-30) — API DEVELOPMENT & DEPLOYMENT

What I Provide:

`src/api.py` (400+ lines, production-quality)

FastAPI application with: - Pydantic models (TransactionInput, PredictionResponse) - `/predict` endpoint (single transaction,

returns probability + decision) - `/batch-predict` endpoint (CSV upload, returns predictions) - `/health` endpoint (model status, inference latency) - `/metrics` endpoint (AUC-PR, precision, recall, F1) - `/threshold` endpoint (adjust decision boundary, recalculate metrics) - Request/response logging (JSON format) - Error handling (invalid inputs, model load failures) - Unit tests (pytest, 10+ test cases) - Request validation, timeout handling

`Dockerfile` - Python 3.11 slim image - Dependency installation - Expose port 8000 - Health check

`railway.yaml` - Railway.app deployment config - Environment variables for MLflow artifact store - Resource limits (memory, CPU)

`requirements-api.txt` - FastAPI, uvicorn, pydantic, httpx, pytest, etc.

`README-API.md` - API endpoints (full specification) - Installation & local testing - Deployment instructions (Railway, Render, AWS alternatives) - Postman collection link - 20 example curl commands

Postman Collection (JSON) - 10 test scenarios: - Single prediction (fraud) - Single prediction (legitimate) - Batch prediction (CSV) - Invalid input handling - Threshold adjustment - Health check - Metrics endpoint - Edge cases (missing fields, wrong types, etc.)

`.gitignore` - `.env` (secrets) - `mlruns/`, `__pycache__/`, `.pytest_cache/` - Kaggle dataset files

What You Do:

Task	Time	Output
Review <code>api.py</code> (understand endpoints, test structure)	30 min	Notes on any feature requests
Set up local FastAPI environment	15 min	Virtual env, dependencies installed
Run pytest locally: <code>pytest src/api.py</code>	15 min	All tests passing ✓
Test endpoints manually (curl or Postman)	30 min	Screenshot of working / predict, /health endpoints
Deploy to Railway/ Render (follow README-API.md)	45 min	Public API URL obtained, working remotely
Write Chapter 4 (System Architecture)	3 hours	1200-1500 words: API design, deployment rationale, diagram
Week 12-13 Total	5.25 hours	Production API deployed, working at public URL

WEEK 14-21 (Nov 6 - Dec 24) — WRITING SPRINT (54 HOURS)

What I Provide:

Outlines & templates for each chapter (in markdown) - Ch.2: Literature Review structure (resampling taxonomy → cost-sensitive

→ ensembles → evaluation theory) - Ch.5: Results structure (technique comparison table → statistical testing → latency results) - Ch.6: Discussion template (publication bias, practical trade-offs, deployment constraints) - Ch.7: Conclusion template (contributions mapped to lit review gaps)

Bibliography (BibTeX exported from Zotero) - 50+ citations formatted, ready to cite

What You Do:

Chapter	Focus	Hours	Words	When
Ch. 1	Introduction: problem framing, fraud losses, why imbalance matters, research question, contribution statement	6	1,200	Week 15-16 (Nov 6-20)
Ch. 2	Literature Review: resampling methods, cost- sensitive learning, ensemble methods, evaluation theory, research gaps	12	2,000	Week 16-17 (Nov 20-27)
Ch. 3	Methodology: dataset (ULB CCFD), features, models, evaluation metrics, CV design, gaps addressed	8	1,500	Week 14-15 (Nov 6-13)
Ch. 4	System Architecture: API, deployment (already written Week 12-13)	3	1,200	Week 12-13 (Oct 16-30)

Chapter	Focus	Hours	Words	When
Ch. 5	Results: technique comparison table, Friedman- Nemenyi test, latency analysis, top techniques, statistical significance	12	2,500	Week 18-19 (Dec 1-11)
Ch. 6	Discussion: why rigorous comparison > novel algorithm, publication bias alert, practical insights, limitations	6	1,200	Week 19-20 (Dec 11-18)
Ch. 7	Conclusion: summary, contributions mapped to gaps, future work, impact statement	4	1,000	Week 21 (Dec 18-24)
Abstract	250-300 words, written LAST	2	300	Week 19
References & Polish	Compile bibliography, fix formatting, final proofread	3	—	Week 20-21 (Dec 18-24)

Chapter	Focus	Hours	Words	When
TOTAL WRITING	Weeks 14-21	54 hours	~10,000 words	Nov 6 - Dec 24

Your Writing Cadence: - Week 14: 8 hrs (Ch. 3 methodology — foundation for results) - Week 15-16: 12 hrs (Ch. 1 + Ch. 2 start — problem framing + lit review) - Week 17: 8 hrs (Ch. 2 completion — taxonomy details) - Week 18-19: 16 hrs (Ch. 5 full + Ch. 6 start — analyze results, discuss implications) - Week 20: 6 hrs (Ch. 6 finish + Ch. 7 + abstract) - Week 21: 4 hrs (Polish, references, final review)

Realistic weekly breakdown: ~6-7 hours/week spread over 8 weeks = not cramming 100 hours in 4 weeks.

WEEK 22-24+ (Dec 24 - End of February 2027) — FINAL REVIEW & SUBMISSION

What I Provide:

- Feedback on any chapters (if you request edits)
- Help with final formatting issues (figure placement, table alignment, citation formatting)

What You Do:

Task	Time	Output
Incorporate supervisor feedback (if any)	2 hours	Revised chapters with notes
Final proofread (spelling, grammar, consistency)	2 hours	Clean, submission-ready PDF
Create title page, declarations, table of contents (per your university template)	1 hour	Properly formatted front matter
Submit to university	30 min	SUBMITTED 
Final Phase Total	5.5 hours	Complete dissertation, submission confirmation

GRAND SUMMARY: Oct 2, 2026 → End of February 2027

Phase	Duration	Your Hours	My Work	Deliverable
Latency & Results	Week 10-11 (Oct 2-16)	5.75 hrs	<code>inference_profiler.py</code> , <code>03_results_analysis.ipynb</code>	Ch. 5 §5.4-5.5 drafted, latency-accuracy plot
API & Deployment	Week 12-13 (Oct 16-30)	5.25 hrs	<code>api.py</code> , Dockerfile, deployment config, tests	Ch. 4 complete, live API at public URL
Writing Sprint	Week 14-21 (Nov 6 - Dec 24)	54 hrs	Chapter outlines, BibTeX templates	7 chapters, abstract, ~10,000 words
Final Review	Week 22-24+ (Dec 24 - Feb 28)	5.5 hrs	Feedback & support	Polished, submitted dissertation
TOTAL	23 weeks	70.5 hours	~1,500 lines production code + 20+ hours design/architecture	Complete MSc Dissertation

CRITICAL NOTES

What's NOT included (out of scope): - Novel algorithm development (you're evaluating existing 12 techniques) - Concept drift analysis (mentioned as "future work" in Ch. 7) - Explainability deep-dive (basic SHAP in notebook; detailed in Ch. 5 §5.5 only) - Advanced MLOps (basic MLflow tracking, not full pipeline orchestration)

Your success depends on: 1. **Following the timeline exactly** — no scope creep, no descoping 2. **Writing incrementally** (200-300 words/day, not 1000-word bursts) 3. **Running code immediately** (don't accumulate "catch up" work) 4. **Emailing supervisor by Oct 15** with scope confirmation (use template from DISSERTATIONCOMPLETEGUIDE.md)

Buffer built in: - You have ~156 hours available (30 hours/week × 5+ months) - Plan allocates 70.5 hours - **85.5-hour buffer (54%)** for unexpected work, supervisor feedback, revisions

NEXT IMMEDIATE STEPS (This Week, Oct 2-8)

1.  Read this breakdown (5 min)
2.  Email supervisor confirming scope (use template from DISSERTATIONCOMPLETEGUIDE.md) — DO THIS TODAY
3.  Pull latest code from your project repo
4.  Verify all Week 1-9 code runs: `python -m pytest src/` (15 min)
5.  Start Week 10: Review `inference_profiler.py` (I'll provide by Oct 4)

By Oct 16: Latency results + Ch. 5 §5.4-5.5 drafted

By Oct 30: API deployed + Ch. 4 complete

By Dec 24: All 7 chapters written

By End of Feb: SUBMITTED 

You're on track. The code is 90% my responsibility; the writing is 100% yours (with my structural support). Let's execute.

Document: RESPONSIBILITIES_BREAKDOWN.md

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Status: Ready for conversion to PDF